

Math Class 9th Federal Board

Chapter 09 Geometry and

Polygons- MCQs , Short and Long

Similar Figures	[SLO: M-09-B-16]: Identify similarity of polygons. Area and Volume of Similar Figures	Summative
	[SLO: M-09-B-17]: Solve problems using the relationship between areas of similar figures and volume of different solids	Summative
Geometrical Properties of regular	[SLO: M-09-B-18]: Solve real life problems that involve the properties of regular polygons, triangles	Summative

Polygons, Triangles and Parallelograms	and parallelograms (such as building architectural structures, fencing, tiling, painting, carpeting a room).	
Loci	[SLO: M-09-B-19]: Solve real life problems using the following loci and the method of intersecting loci for sets of points in two dimensions which are: at a given distance from a given point, at a given distance from a given straight line, equidistant from two given points equidistant from two given intersecting straight lines.	Summative

1. Encircle the correct option in the following.

(i) Which of the following is polygon?

(a) circle

(b) pyramid

(c) quadrilateral

(d) sphere

(ii) Which of the following is regular polygon?

(a) kite

(b) rhombus

(c) rectangle

(d) square

(iii) If two triangles are similar, their corresponding sides are.

(a) proportional

(b) equal

(c) congruent

(d) parallel

(iv) What is the sum of interior angles for an irregular hexagon?

(a) 120°

(b) 720°

(c) 135°

(d) 360°

(v) What is the sum of interior angles for a regular 12 sided polygon?

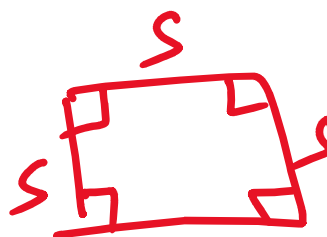
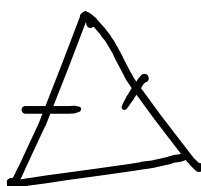
(a) 1800°

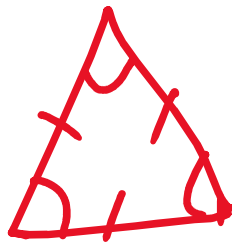
(b) 2160°

(c) 1980°

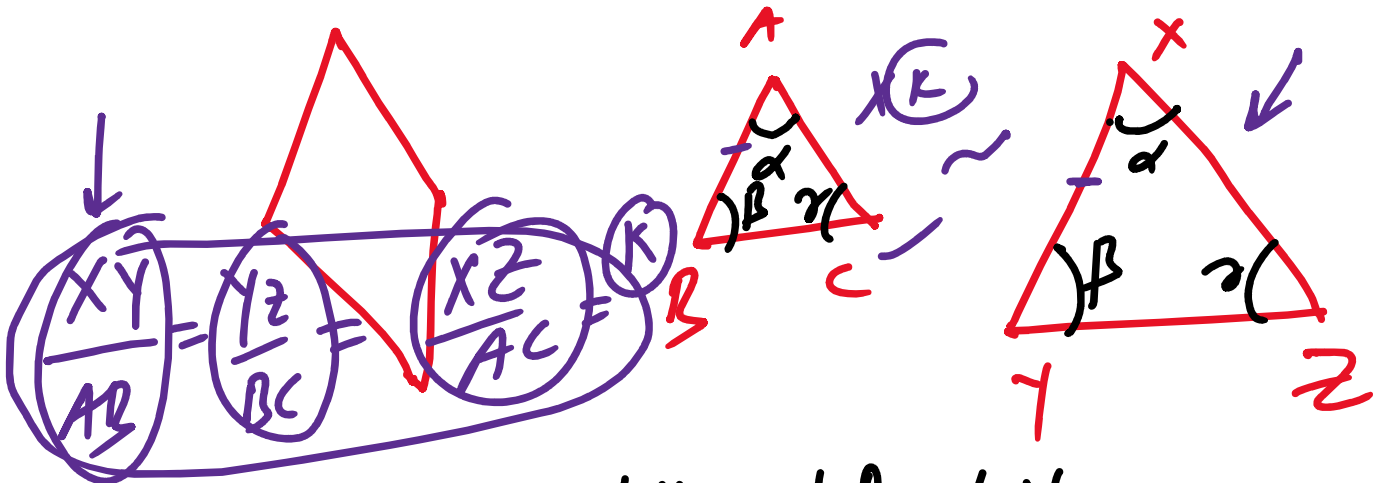
(d) 360°

i)





Regular polygon
 Sides are equal
 internal Angles are Equal



proportion $\angle A = \angle X$, $\angle B = \angle Y$
 $\angle C = \angle Z$

(iv) Hexagon (Regular or Irregular)

Total sum of Interior Angles



$$= (n-2) \times 180^\circ$$

$$= (6-2) \times 180^\circ =$$

$$\rightarrow = (6-2) \times 180^\circ =$$

Interior angle of a Regular Hexagon

$$\begin{aligned} n=6 &= \frac{(n-2) \times 180^\circ}{n} \rightarrow \text{Total Interior angle} \\ &= \frac{4 \times 180^\circ}{6} = 120^\circ \end{aligned}$$

(v) What is the sum of interior angles for a regular 12 sided polygon?

(a) 1800°

(b) 2160°

(c) 1980°

(d) 360°

$$= (n-2) \times 180^\circ$$

$$= \underline{(12-2)} \times 180^\circ = 1800^\circ$$

Each interior angle
of Regular 12-sided
poly -

$$\begin{aligned} &\frac{1800}{12} \\ &= 150^\circ \end{aligned}$$

(vi) How many sides a regular polygon has if its exterior angle is 15°

(a) 20

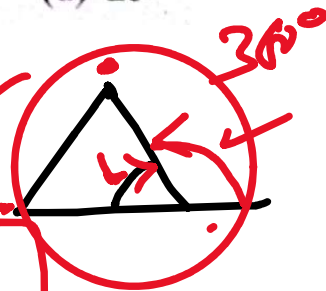
(b) 21

(c) 24

(d) 25

$$\frac{360^\circ}{15^\circ} = 24$$

Exterior angle of a regular polygon



$$\Rightarrow \boxed{\text{Ex. angle} = \frac{360^\circ}{n}}$$

$$15^\circ = \frac{360^\circ}{n}$$

$$n = \frac{360^\circ}{15} = 24 \text{ sides}$$



$$\text{Interior angle} = 180^\circ - \text{Exterior angle}$$

$$\text{Exterior angle} = 180^\circ - \text{Interior angle}$$

$$\text{Interior angle} \dots \dots \dots \text{rule} = 180^\circ$$

Interior + Exterior Angle = 180

21 3 1 part

(a) ~~20~~ (b) ~~21~~

(c) ~~24~~ 9cm (d) ~~25~~ 12cm

(vii) In the figure, $AB = 21$ cm and

A P B

P divides AB in the ratio 3 : 4. What is length of $\overline{AP}$?

(a) 8 cm

(b) 10 cm

(c) 12 cm

(d) 9 cm

$$\overline{AP} = \frac{3}{7} \times 21 = 9 \text{ cm}$$

$$\overline{BP} = \frac{4}{7} \times 21 = 4 \times 3 = 12 \text{ cm}$$

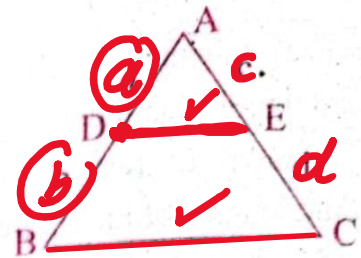
(viii) In the figure, $\frac{a}{b} = \frac{c}{d}$. Which one is true?

(a) $\overline{DE} = \overline{BC}$

(b) $\overline{DE} > \overline{BC}$

(c) $\overline{DE} \cong \overline{BC}$

(d) $\overline{DE} \parallel \overline{BC}$



Equal or
Congruent ($\cong$)

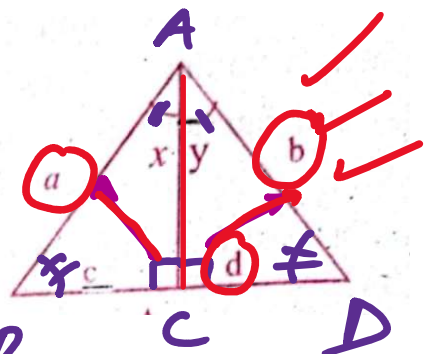
(ix) In the figure if $x = y$. Then the value of b is:

(a) $\frac{ac}{d}$

(b) $\frac{ad}{c}$

(c) $\frac{cd}{a}$

(d) $\frac{c}{cd}$



$$\angle B = \angle D$$

(b)

d

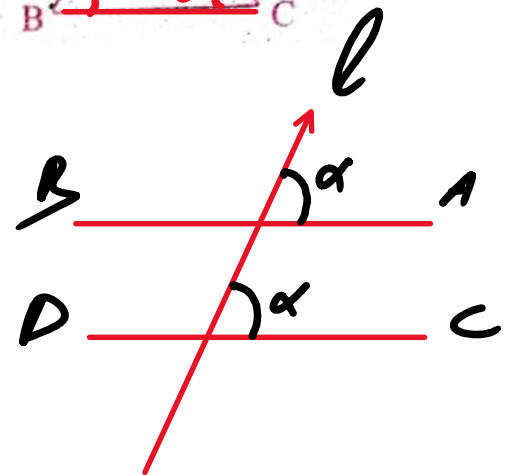
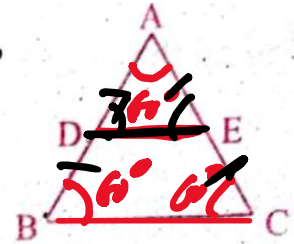
$\triangle ABC \sim \triangle ACD$

$$\frac{b}{a} = \frac{d}{c} \quad \triangle ABC \sim \triangle ACD$$

$$b = \frac{ad}{c}$$

(x) In the figure, $\triangle ABC$ is equilateral and $\overline{DE} \parallel \overline{BC}$, then $\triangle ADE$ is

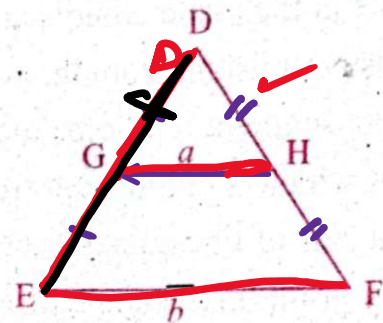
- (a) right angled
(b) scalene
(c) isosceles
(d) equilateral



(xi) In the figure, $\overline{GH} \parallel \overline{EF}$.

Then $a : b = ?$

- (a) $DG : DE$
(b) $DG : DH$
(c) $DG : GE$
(d) $DE : EF$



$$a : b \Rightarrow \frac{GH}{EF}, \quad \triangle DGH \sim \triangle DEF$$

$$\frac{GH}{EF} = \frac{DG}{DE} = \frac{DH}{DF} = \frac{a}{b} \quad \text{similar}$$

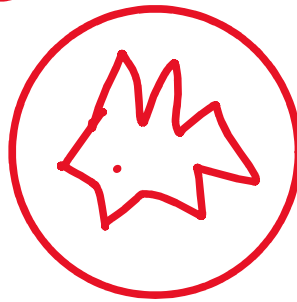
(xii) What is the sum of all the exterior angles of a 13-sided polygon whose one interior angle is equal to x° ?

(a) $90^\circ + x$

(b) 360°

(c) $360^\circ + x$

(d) $180^\circ + x$



360°

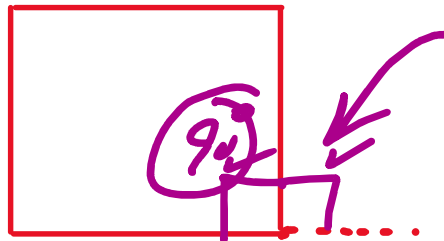
(xiii) Which polygon has both its interior and exterior angles the same?

(a) pentagon

(b) triangle

(c) square

(d) hexagon



(xvii) Two solids with equal ratios of corresponding linear measures:

(a) are similar

(b) are congruent

(c) have different area

(d) have same volume

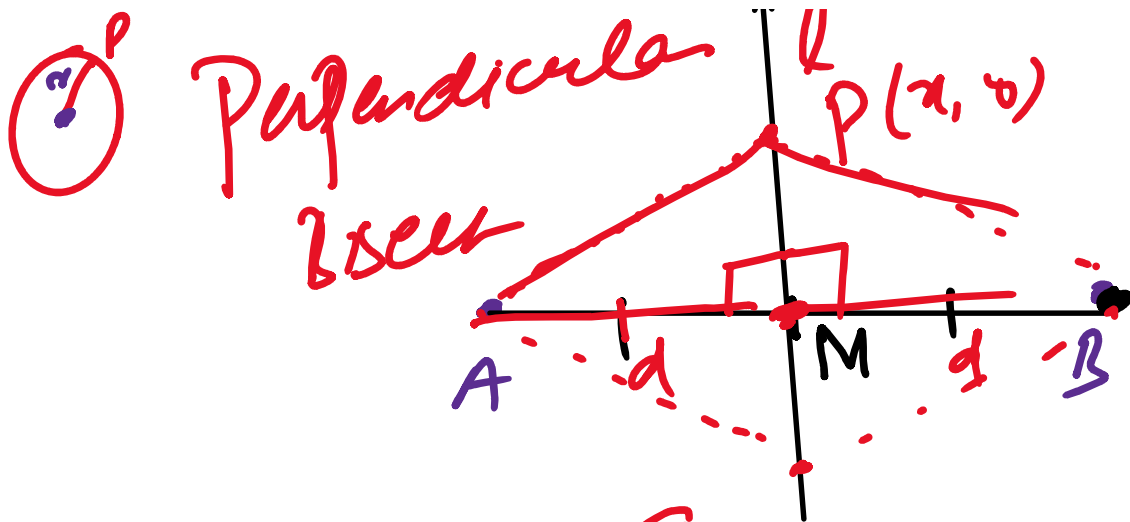
$\frac{A_1}{A_2}$

$\frac{V_1}{V_2}$

$\frac{l_1}{l_2} = \frac{w_1}{w_2} = \frac{h_1}{h_2}$

The locus of points equidistant from two given points is the Perpendicular bisector

Perpendicular bisector

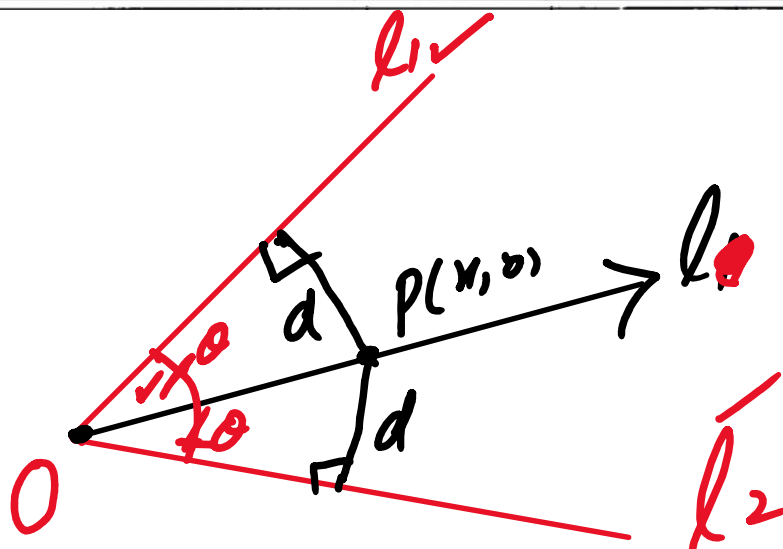


11. Each interior angle of a regular square-shaped garden is:	90°	60°	45°	30°
12. The locus of points equidistant from two intersecting straight lines is formed by the:	Two parallel lines	Perpendicular bisectors	Circles	Angle bisectors

Interior Angle = $\frac{(n-2) \times 180^\circ}{n}$

$= \frac{(4-2) \times 180}{4} = 90^\circ$

11. Each interior angle of a regular square-shaped garden is:	90°	60°	45°	30°
12. The locus of points equidistant from two intersecting straight lines is formed by the:	Two parallel lines	Perpendicular bisectors	Circles	Angle bisectors



If two polygons are similar in shape and ratio

9.	If two polygons are similar in shape and ratio of their corresponding sides is 2:3, what is the ratio of their areas? اگر دو متماثل کثیر پہلو اشکال کے متقابلہ اضلاع میں نسبت 2:3 ہو تو ان کے رقبوں میں کیا نسبت ہوگی؟	2:3	4:9	8:27	1:1
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$$\frac{l_1}{l_2} = \frac{2}{3}$$

$$\Rightarrow \frac{A_1}{A_2} = \left(\frac{2}{3}\right)^2 = \frac{4}{9} = \frac{4 \times 1}{2 \times 2}$$

$$\frac{V_1}{V_2} = \left(\frac{2}{3}\right)^3 = 8:27$$

9.	What is the value of x in the given similar triangles? دی گئی متماثل مثلثان میں x کی ویلیو کیا ہے؟	1	2	3	6
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$$\frac{x}{6} \rightarrow \frac{3}{9} \Rightarrow x = \frac{2 \times 1}{8 \times 3}$$

$$x = 2$$

$$x = 8$$

11.	Each interior angle of a regular octagon is equal to: ایک باقاعدہ ہشت پہلو شکل کا ہر اندرونی زاویہ _____ ہوتا ہے۔	108°	120°	135°	140°
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$$\Rightarrow (n-2) \times 180^\circ = (8-2) \times 180^\circ$$

$$\Rightarrow \frac{(n-2) \times 180^\circ}{n} = \frac{(8-2) \times 180^\circ}{8}$$

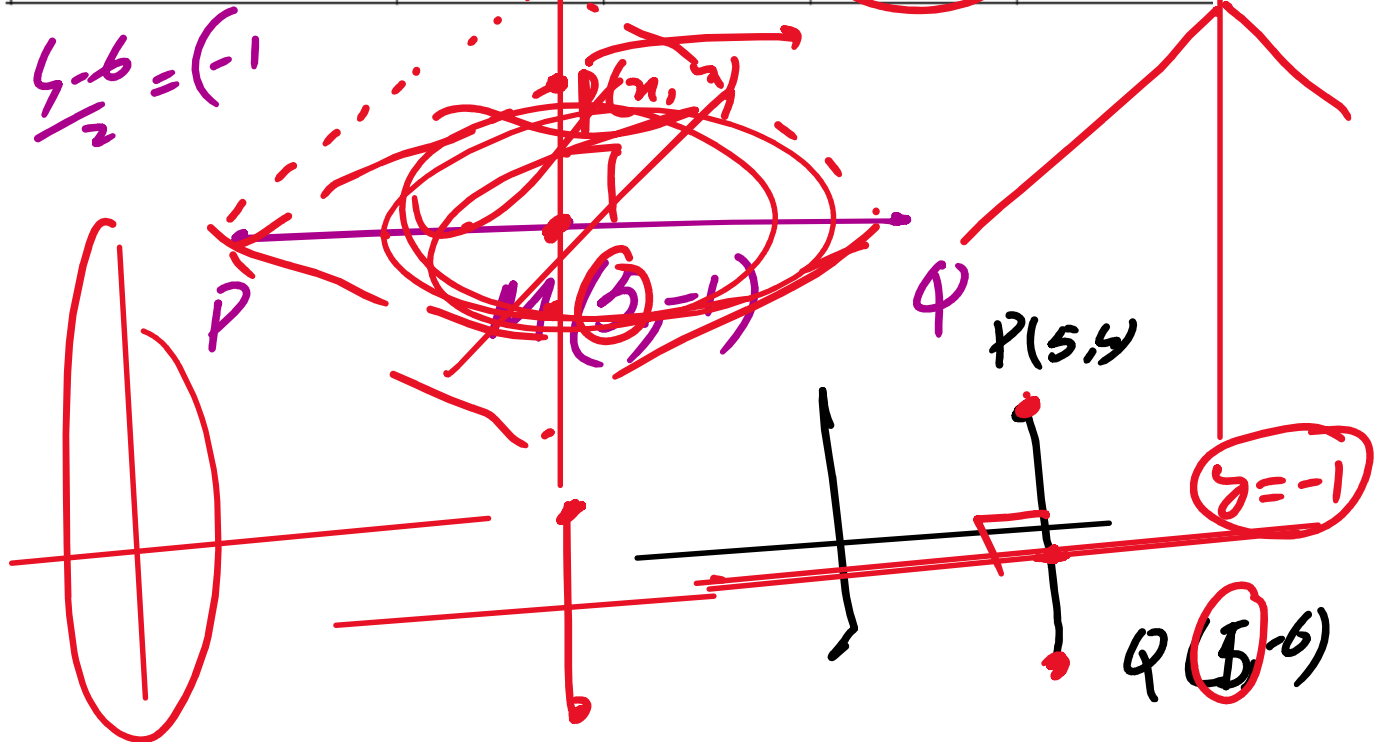
$$= \frac{6 \times 180^\circ}{8} =$$

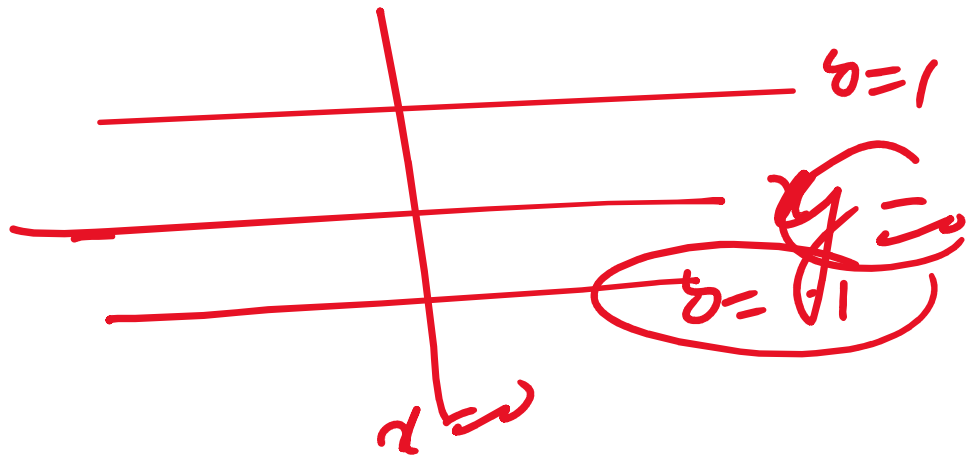
2nd method

$$\text{Ex. Angle} = \frac{360^\circ}{8} = 45^\circ$$

$$\text{Interior angle} = 180^\circ - 45^\circ = 135^\circ$$

Locus of points equidistant from $P(5,4)$ and $Q(5,-6)$ is:	$x = 0$	$x = 5$	$y = -1$	$y = 1$
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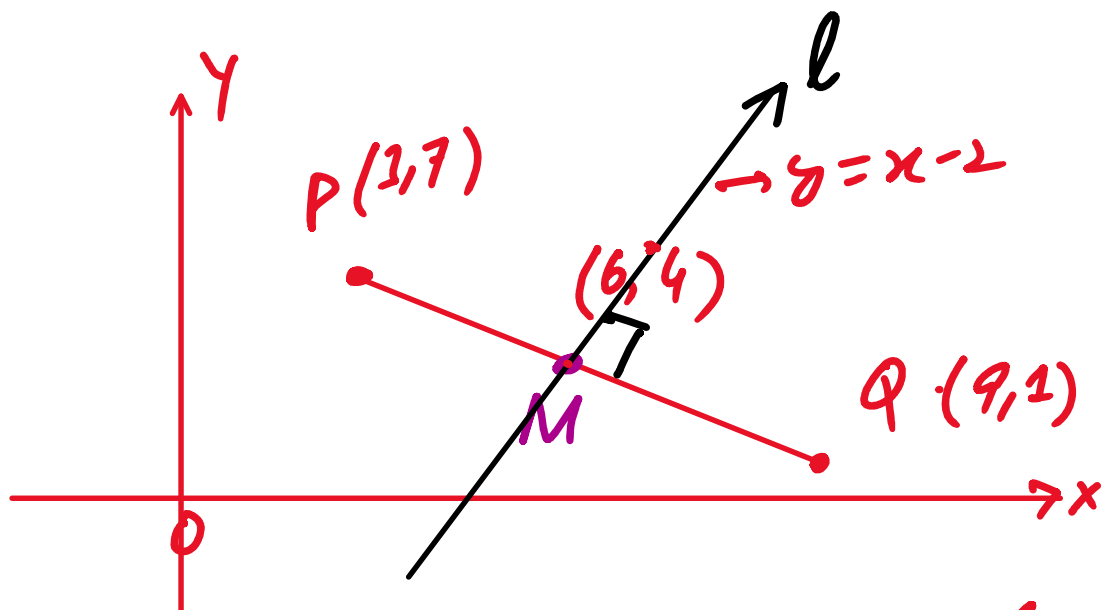




Short Questions

(vii)	A water fountain is to be placed at a point which is equidistant from two lamp posts located at $P(3,7)$ and $Q(9,1)$. Find possible positions of the fountain.	<u>3+1</u>
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Sol:



The water fountain will lie on the perpendicular bisector of line seg. $\overline{PQ}$

Ans. 37 1.4
Mid point of $\overline{PQ} = M(x, y)$

$$x = \frac{3+9}{2} = \frac{12}{2} = 6$$

$$y = \frac{7+1}{2} = \frac{8}{2} = 4$$

So, $M(6, 4)$ is the mid-point of $\overline{PQ}$

Slope of line $\overline{PQ} \Rightarrow m_1 = \frac{1-7}{9-3}$

$$m_1 = \frac{-6}{6} \Rightarrow \boxed{m_1 = -1}$$

As L bisects line $\overline{PQ}$ is perpendicular to $\overline{PQ}$ so, its slope

$$\boxed{m_2 = 1} \quad (\because m_1 \times m_2 = -1)$$

Eqn. of line $\overline{L}$ using point-slope form

29. 7 points 1' form

$$y - y_1 = m(x - x_1)$$

$$m_2 = 1, M(\overset{x_1, y_1}{6, 4})$$

$$\Rightarrow y - 4 = 1(x - 6)$$

$\Rightarrow y - 4 = x - 6$

$$x - y + 4 - 6 = 0$$

$$\boxed{x - y - 2 = 0}$$

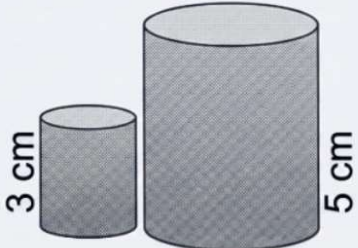
$$\Rightarrow \boxed{y - x - 2}$$

is the required line where
fountain can be placed
on this line.

anywhere on this line.

Possible positions

There are infinitely many possible positions of the lamp -

(viii)	In the figure two geometrically similar cylinders are shown. Find:		
	a. The ratio of <u>volume</u> of <u>smaller cylinder</u> to <u>larger cylinder</u>. b. Curved surface area of <u>smaller cylinder</u> if that of larger one is 250cm^2. c. Volume of larger cylinder if volume of smaller one is 162cm^3.		2+1 +1

Sol: (a) Let V_s and V_L be the volume of smaller & larger cylinders respectively.

Given $k = \underline{3}$ ✓

Given $\frac{l_1}{l_2} = \frac{3}{5} \checkmark$

$$\Rightarrow \frac{V_{s'}}{V_L} = \left(\frac{3}{5}\right)^3 = \frac{27}{125}$$

$$\Rightarrow \boxed{V_s : V_L = 27 : 125} \checkmark$$

(b) $\frac{A_s}{A_L} = \left(\frac{3}{5}\right)^2$

$$\Rightarrow A_s = \frac{9}{25} \times 250 \text{ cm}^2$$

$$\Rightarrow \boxed{A_s = 90 \text{ cm}^2} \checkmark$$

(c) $V_s = 162 \text{ cm}^3$

$$\Rightarrow \checkmark A_s \frac{V_s}{11} = \frac{27}{125}$$

$$\therefore \frac{V_L}{V_S} = \frac{125}{27}$$

$$\Rightarrow \frac{V_L}{V_S} = \frac{125}{27}$$

$$\Rightarrow V_L = \frac{125}{27} \times V_S$$

$$V_L = \frac{125}{27} \times 162 \text{ cm}^3$$

$$\boxed{V_L = 750 \text{ cm}^3} \quad \text{Ans}$$

The ratio of areas of the bases of two similar cylinders is 9:4. The area of base of the smaller cylinder is 240 cm^2 . Calculate the radius, diameter and area of the base of large cylinder.



Sol:- Given, Let A_L and A_S be the base areas of cylinders

$$A_L : A_S = 9 : 4$$

$$\Rightarrow \frac{\sqrt{A_L}}{A_S} = \frac{9}{4} \rightarrow \textcircled{1}$$

$$A_S = 240 \text{ cm}^2$$

$$\textcircled{1} \Rightarrow A_L = \frac{9}{4} \times A_S$$

$$A_L = \frac{9}{4} \times 240 \text{ cm}^2$$

$$\boxed{A_L = 540 \text{ cm}^2}$$

As we know the Base Area = πr^2

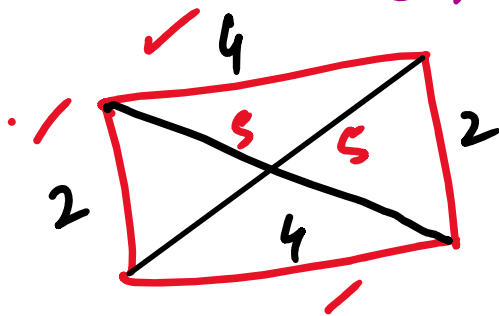
$$\Rightarrow \pi r^2 = 540 \text{ cm}^2$$

$$r = \sqrt{\frac{540}{\pi}} \text{ cm}$$

$$r = 13.11 \text{ cm}$$

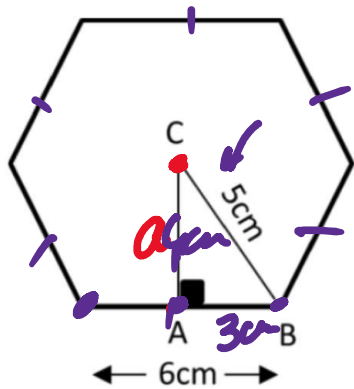
As Diameter = $2r$

$$\Rightarrow d = 26.22 \text{ cm}$$



$$\text{perimeter} = 2(4+2) = 2 \times 6 = 12 \text{ cm}$$

In the given figure, find area of a regular hexagonal roof of a building shown below.



Sol:

$$\text{Area} = \frac{1}{2} \times P \times a \rightarrow \textcircled{1}$$

where

$P = \text{perimeter of regular Hexagon}$

and $a = \text{Apothem} = CA$

$$P = 6 \times 6 = 36 \text{ cm}$$

11

$$AB = \frac{6\text{cm}}{2} = 3\text{cm}$$

By Pythagorean Theorem

$$5^2 = 3^2 + a^2$$

$$\Rightarrow a = \sqrt{25 - 9}$$

$$a = \sqrt{16} = 4\text{cm}$$

$$\Rightarrow \boxed{a = 4\text{cm}}$$

①
 $\Rightarrow$

$$\text{Area} = \frac{1}{2} \times 36 \times 4\text{cm}^2$$

$$\boxed{\text{Area} = 72\text{cm}^2} \quad \text{A}$$

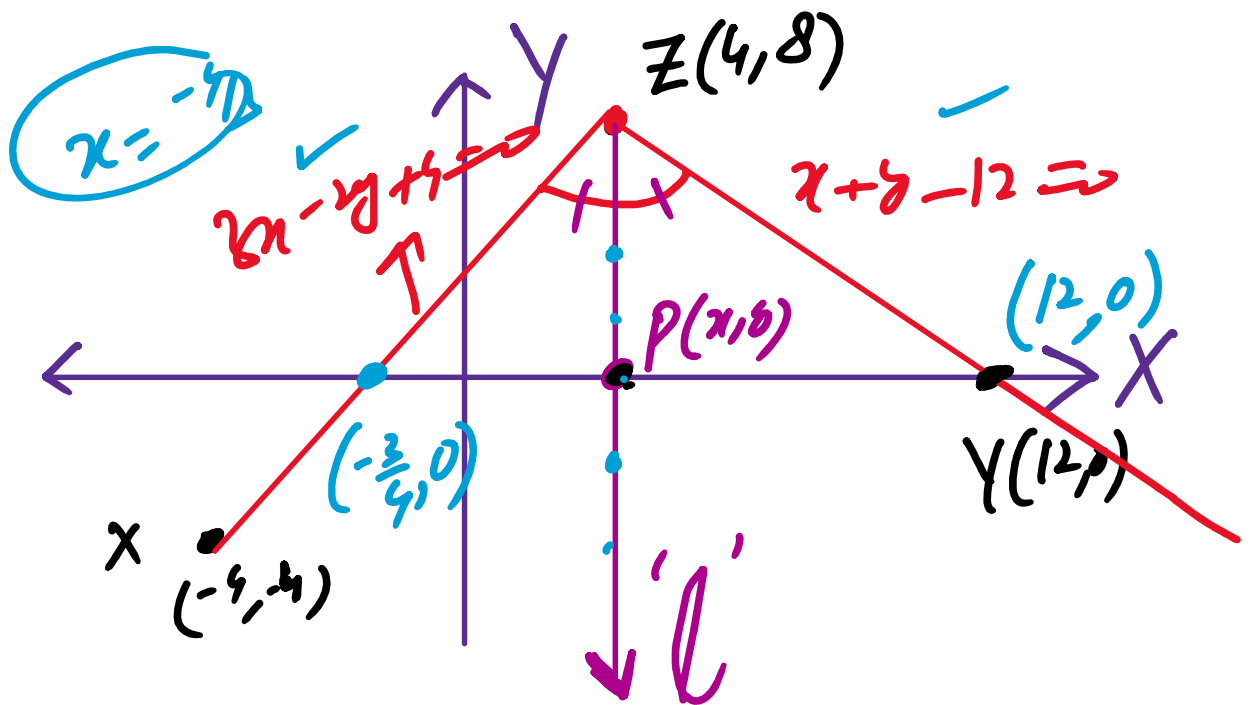
A triangular garden XYZ shows corners $X(-4, -4)$, $Y(12, 0)$ and $Z(4, 8)$ geometrically. Find locus of the corners equidistant from XZ and YZ .

4

line line

Sol:

The foci of conics
lie at the angle bisector
of XZ and YZ



Eqn. 7 $\bar{XZ}$

7

slope of $XZ = \frac{8+9}{4+4} = \frac{12}{8} = \frac{3}{2} = m_1$

✓ 1 7 411 ✓
As $X(-4, -4)$ lies on XZ line

By using point-slope form

$$y - y_1 = m(x - x_1)$$

$$y + 4 = \frac{3}{2}(x + 4)$$

$$2y + 8 = 3x + 6$$

$$\boxed{3x - 2y + 4 = 0}$$

Eqn. of XZ .

For Eqn. of YZ

$$m_2 = \text{slope of } YZ = \frac{8-0}{4-12} = \frac{8}{-8} = -1 \quad \text{--- (1)}$$

$$m_2 = -1$$

$$Y(12, 8)$$

$$y - 0 = -1(x - 12)$$

$$y = -x + 12$$

$$\boxed{x + y - 12 = 0}$$

Eqn. of YZ

Let $P(x, y)$ be any point
lies on Angle Bisector of
 XZ & YZ & it is
equidistant from both lines.

$$d = \frac{|3x - 2y + 4|}{\sqrt{9 + 4}} = \frac{|x + y - 12|}{\sqrt{1 + 1}}$$

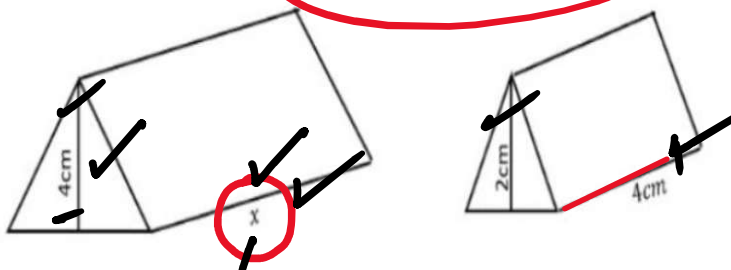
$$\boxed{|3x - 2y + 4| + |x + y - 12|}$$

$$= \left[\frac{3x - 2y + 4}{\sqrt{13}} = \pm \frac{x + y - 4}{\sqrt{2}} \right]$$

is the required locus of garden corner.

There are infinitely many points

In the adjacent similar figures,



find the value of x and the ratio of volumes v_1 and v_2 .

Sol:-

$$\frac{x}{4} = \frac{4}{2} = 2 \text{ cm}$$

$$x = 8 \text{ cm}$$

As $\frac{h_1}{h_2} = \frac{4}{2} = \frac{8}{4} = \frac{2}{1}$

$$\Rightarrow h_1 : h_2 = 2 : 1$$

$$\Rightarrow \mu : x_2 - x_1 \dots$$

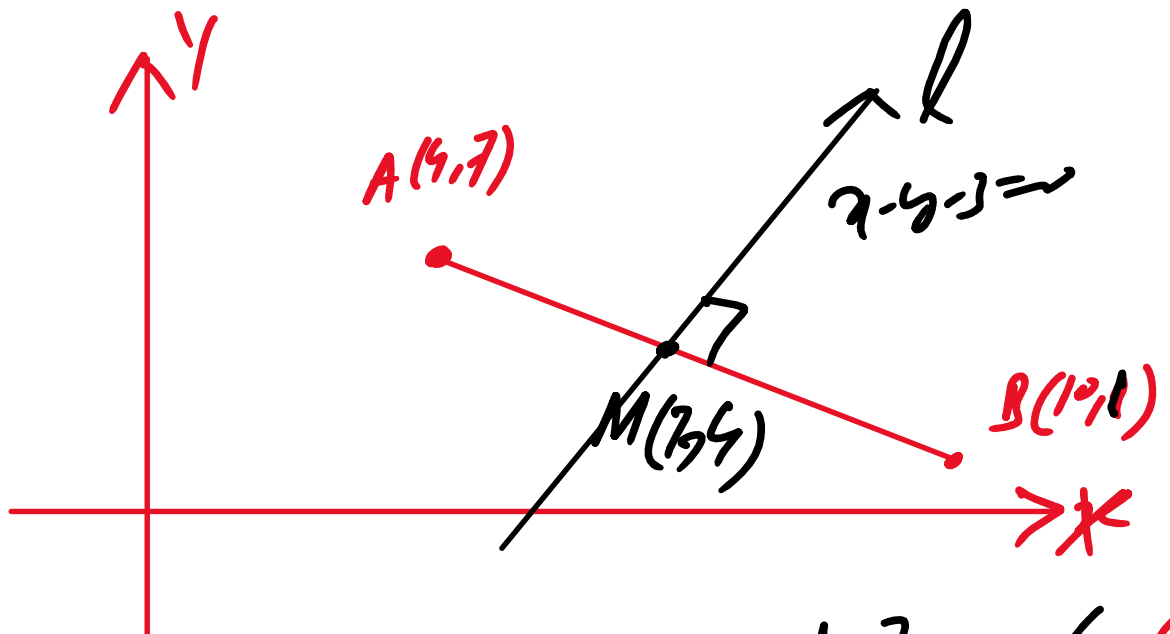
$$\frac{V_1}{V_2} = \left(\frac{r_1}{r_2} \right)^3 = \left(\frac{2}{1} \right)^3 = \frac{8}{1}$$

$$\Rightarrow \boxed{V_1 : V_2 = 8 : 1} \neq$$

A city is planning to install a new water distribution hub such that it is equidistant from two existing water stations located at $A(4,7)$ and $B(10,1)$. Find the equation of the locus where the new hub should be placed.

04

ایک شہر میں پانی کی تقسیم کا ایک نیا مرکز بنانے کا منصوبہ ہے۔ انتظامیہ چاہتی ہے کہ نیا منصوبہ کے پہلے سے نقاط $A(4,7)$ اور $B(10,1)$ پر گئے مراکز سے مساوی فاصلہ پر ہو۔ اس لوکس کی مساوات معلوم کریں جہاں نیا منصوبہ لگنا چاہیے۔



$$\text{slope of } AB = m_1 = \frac{1-7}{10-4} = \frac{-6}{6} = -1$$

$$\text{slope of } AB = m_1 = \frac{7-4}{10-9} = 3$$

$$\text{slope of Line Bisector } l' = 1 = m_2$$

$$\begin{aligned} \text{Mid-point of } AB &= M\left(\frac{9+10}{2}, \frac{7+4}{2}\right) \\ &= M(7, 4) \checkmark \end{aligned}$$

Use point-slope

$$y - 4 = 1(x - 7)$$

$$y - 4 = x - 7$$

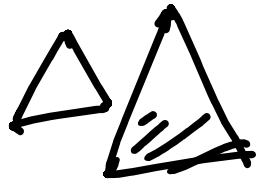
$$\boxed{x - y - 3 = 0}$$

is the required locus where
the dis. has can be placed.

⇒ There are infinitely many points

→ These are important problems
No Similar figures

A cylindrical water tank has a radius of 5 cm and a height of 12 cm. A second cylindrical tank has a radius of 7.5 cm. Find the height of the second tank if both tanks have the same volume.



Sol: $r_1 = 5 \text{ cm}, h_1 = 12 \text{ cm}$

$r_2 = 7.5 \text{ cm}, h_2 = ?$

Volume = Equal.

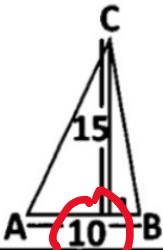
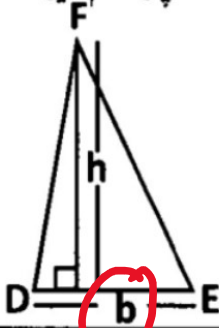
$V_1 = V_2$ given

$\pi r_1^2 h_1 = \pi r_2^2 h_2$

$h_2 = \frac{r_1^2 h_1}{r_2^2}$

$h_2 \frac{(5)^2 \times 12}{(7.5)^2} \Rightarrow h_2 = \frac{25 \times 12}{56.25} =$

$$h_2 = 5.33 \text{ cm}$$

(viii)	Two similar triangles have areas in the ratio 25:49. The base and height of the smaller triangle are 10 cm and 15 cm, respectively. Find the corresponding base and height of the larger triangle.	دو متماثل مثلثان کے رقبوں کی نسبت 25:49 ہے۔ اگر ایک مثلث کا قاعدہ 10 سم اور اس کی اونچائی 15 سم ہوں تو دوسرے مثلث کا قاعدہ اور اونچائی معلوم کریں۔	04
			

Sol:

Given

$$A_s : A_L = 25 : 49$$

$$\Rightarrow \frac{A_s}{A_L} = \frac{25}{49} = \left(\frac{l_s}{l_L} \right)^2$$

$$\Rightarrow \sqrt{\left(\frac{l_s}{l_L} \right)^2} = \sqrt{\frac{25}{49}}$$

✓ (✓) ✓

$$\Rightarrow \boxed{\frac{l_s}{l_L} = \frac{5}{7}} \checkmark$$

Also, $b_s = 10 \text{ cm} \checkmark$

$h_s = 15 \text{ cm} \checkmark$

$b_L = ? \quad h_L = ?$

$\Rightarrow \frac{b_s}{b_L} = \frac{5}{7} \checkmark$

$\Rightarrow \frac{10}{b_L} \neq \frac{5}{7}$

$\Rightarrow 5b_L = 70$

$b_L = \frac{70}{5} \text{ cm}$



$$\boxed{b_L = 14 \text{ cm}} \quad \checkmark$$

Similarly $\frac{h_L}{h} = \frac{5}{7} \checkmark$

$$\frac{h_L}{15} = \frac{7}{5} \Rightarrow h_L = \frac{7 \times 15}{5}$$

$$\boxed{h_L = 21 \text{ cm}} \quad \checkmark$$

A regular hexagon has a side of 6 cm. Find its perimeter and each interior angle.

ایک باقاعدہ چھ ضلعی شکل (مسدس) کی ہر ضلع 6 سینٹی میٹر ہے۔ اس کا محیط (perimeter) 04
اور ہر اندرونی زاویہ معلوم کریں۔

Sol.

$$\text{Side} = 6 \text{ cm}$$

7

Perimeter = ?
Each Interior Angle = ?

As $n = 6$ ✓

$$\text{Perimeter} = 6 \times 6 = \underline{\underline{n \times s^{\text{cm}}}}$$

$$= 36 \text{ cm}$$

$$\text{Each Interior Angle} = \frac{(6-2) \times 180^\circ}{6}$$

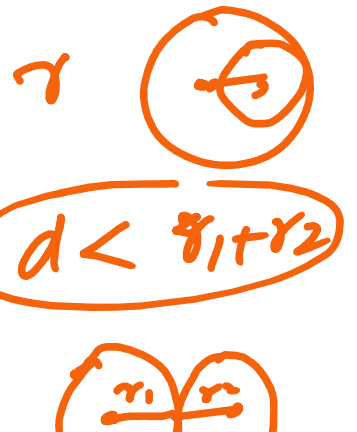
$$= \frac{4 \times 180^\circ}{6} = 120^\circ$$

(vii)

There are two trees 50 feet apart. It is planned to place a lamp post at 35 feet distance from each tree. Draw a sketch to show where the lamp-post can be placed in relation to the trees. How many locations for the lamp post are possible?

دو درخت باہمی 50 فٹ کے فاصلے پر ہیں۔ ایک منصوبہ کے مطابق ایک لیپ پوسٹ ایسے لگانا مقصود ہے کہ اس کا فاصلہ ہر درخت سے 35 فٹ ہو۔ ایک سکیچ بنا کر لیپ پوسٹ کی ممکنہ جگہ دکھائیں۔ لیپ پوسٹ کتنی جگہوں پر لگایا جاسکتا ہے؟

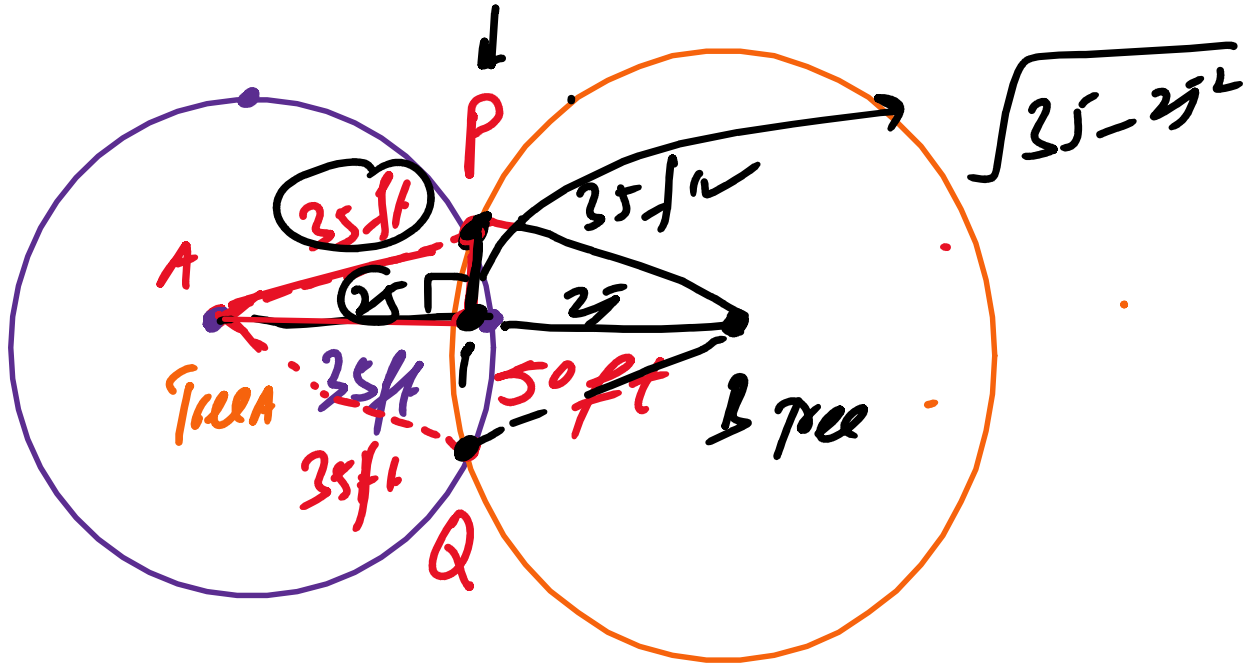
04



مقصود ہے کہ اس کا فاصلہ ہر درخت سے 35 فٹ ہو۔ ایک بیچ بنا کر لیپ پوسٹ کی ملنے جگہ دکھائیں۔ لیپ پوسٹ کتنی جگہوں پر لگایا جاسکتا ہے؟

$$\begin{aligned} & \text{Diagram showing two overlapping circles with centers } r_1 \text{ and } r_2. \\ & r_1 + r_2 = 70 \end{aligned}$$

Sol:



The lamp-post must be placed at the intersection points P & Q of both circles having centres at two trees A and B, having radius of 35 feet

radius of 35 feet

Possible fixations = 2

(viii)	Two similar triangles have areas 36 cm^2 and 64 cm^2 . Find ratio of the corresponding sides of the triangles if one side of the smaller triangle is 6 cm. دو مشابہ مثلثوں کے رقبے 36 cm^2 اور 64 cm^2 ہیں اگر چھوٹی مثلث کا ایک ضلع 6 cm ہو تو ان کی اطراف کا تناسب معلوم کریں۔	04
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سول

$$\frac{A_s}{A_L} = \frac{36}{64}$$

$$\frac{l_s}{l_L} = \sqrt{\frac{36}{64}}$$

$$\frac{l_s}{l_L} = \frac{6}{8} = \frac{3}{4}$$

$$\boxed{\frac{l_s}{l_L} = \frac{3}{4}}$$

ratios
of
corresponding
sides

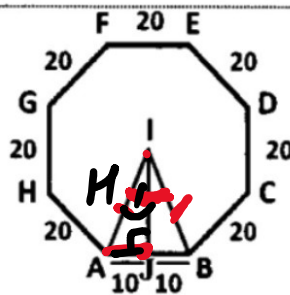
$$\Rightarrow \frac{6}{h} = \frac{3}{4}$$

$$\Rightarrow \frac{12}{6} = \frac{4}{3}$$

$$\Rightarrow R_L = \frac{4 \times 16^2}{8}$$

$RL = 8 \text{ cm}$

دی گئی تناسب ہشت پہلو ABCDEFGH کی ہر
سائیڈ 20 سم ہے۔ درج شدہ معلوم کریں:



- a** The angles $\angle AIB$ and $\angle AIJ$
b The lengths of AI and IJ
c Area of $\triangle AIB$
d Area of the octagonal tile

~~بیش بهای کل کار~~

Sol:-

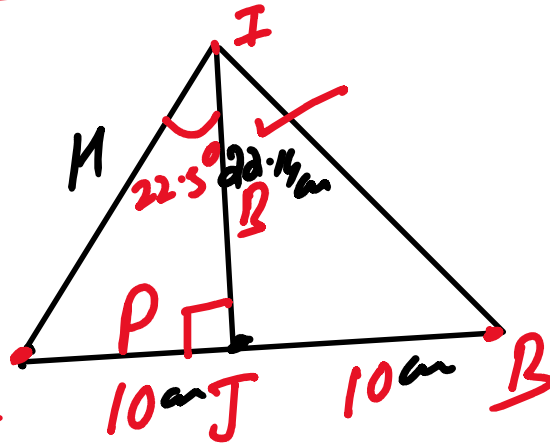
$$LAIB = \frac{36^\circ}{8}$$

08 $\angle A B = 45^\circ$

$$[LAIJ = \frac{45^\circ}{2} = 22.5^\circ] \Rightarrow$$

$$\angle HIJ = \frac{1}{2} = 0.5$$

(b)



In rt. $\triangle AIJ$

$$\tan 22.5^\circ = \frac{10}{IJ}$$

$$\Rightarrow IJ = \frac{10}{\tan 22.5^\circ}$$

$$IJ = 22.14 \text{ m}$$

In rt. $\triangle AIT$

$$\sqrt{AI^2} = \sqrt{AJ^2 + IJ^2}$$

$$AI = \sqrt{10^2 + 22.14^2}$$

$$AI = 24.28 \text{ cm}$$

c) Area of $\triangle AIB$

$$= \frac{1}{2} \times \text{base} \times \text{height}$$

$$= \frac{1}{2} AB \times AI$$

$$\text{Area} = \frac{1}{2} \times 22 \text{ cm} \times 22.14 \text{ cm}$$

$$\boxed{\text{Area} = 221.4 \text{ cm}^2 \text{ of } \triangle AIB} \quad R_2$$

d Area of the octagonal tile

هشت پہلو نائل کا رقبہ

Area of two octagonal tiles

$$= 8 \times \triangle IAB$$

$$= 8 \times 221.4 \text{ cm}^2$$

$$= 8 \times 221.4 \text{ cm}$$

$$\boxed{\text{Area} = 1771.2 \text{ cm}^2}$$

2nd method

Area of octagonal tile

$$= \frac{1}{2} P \times a \rightarrow (1)$$

$$\text{Perimeter} = 8 \times 20 = 160 \text{ cm}$$

$$a = \text{Apothem} = 22.14 \text{ cm}$$

(1) =

$$\text{Area} = \frac{1}{2} \times 160 \text{ cm} \times 22.14 \text{ cm}$$

$$\boxed{\text{Area of tile} = 1771.2 \text{ cm}^2}$$